

Multiple Choice (10 marks)

1. In Python 3, which of the following function removes all leading and trailing whitespace in string?
 - (a) `replace(old, new [, max])`
 - (b) `strip([chars])`
 - (c) `swapcase()`
 - (d) `title()`
2. Which of the following regular expressions is equivalent to (describes the same set of strings as) $(a^* + b)^*(c + d)$?
 - (a) $a^*(c + d) + b(c + d)$
 - (b) $a^*(c + d)^* + b(c + d)^*$
 - (c) $a^*(c + d) + b^*(c + d)$
 - (d) $(a + b)^*c + (a + b)^*d$
3. In Python 3, what is the output of `print list[1:3]` if `list = [ 'abcd', 786 , 2.23, 'john', 70.2 ]`?

786 , 2.23, 'john', 70.2

 - (a) `abcd`

786, 2.23

 - (b) None of the above.
4. What is the output of the statement `"a" + "ab"` in Python 3?
 - (a) Error
 - (b) `aab`
 - (c) `ab`
 - (d) `a ab`
5. Which of the following best explains what happens when a new device is connected to the Internet?
 - (a) A device driver is assigned to the device.
 - (b) An Internet Protocol (IP) address is assigned to the device.
 - (c) A packet number is assigned to the device.
 - (d) A Web site is assigned to the device.

True / False (10 marks)

Answer True or False

6. True or False: The set of operators {AND, OR} is not complete, as it cannot represent all Boolean expressions.
7. True or False: The access matrix approach can be large and cumbersome to manage when stored directly.
8. True or False: A distributed denial-of-service (DDoS) attack involves multiple computers launching the attack simultaneously.
9. True or False: The maximum possible degree of the minimal-degree interpolating polynomial $p(x)$ for $n + 1$ distinct points is n .
10. True or False: A function that averages numeric values in a column or row is effective for detecting outliers from data entry errors.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to extract the maximum numeric value from a string by using regex.
12. Write a function to find the minimum total path sum in the given triangle.

End of Paper